

Bicycle Helmet Compliance vs. Adolescent Interpersonal Violence

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Course: Statistics Practice | Cycle 4
Summary

1. STUDY FRAMEWORK

- Objective: Does proactive regulatory compliance (helmets) link to general risk-mitigation (less fighting)?
- Dataset: YRBS 2007 Nationwide Survey
- X (Explanatory): Helmet_Numeric (1: Always, 0: Never)
- Y (Response): Fight_Numeric (1: Fought, 0: No Fight)

11,144

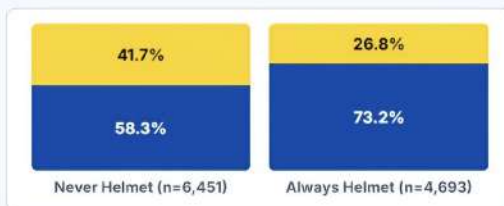
TOTAL VALID OBSERVATIONS (N)

-16.154

Z-STATISTIC (HIGHLY SIGNIFICANT)

2. DESCRIPTIVE & MODEL RESULTS

- Proportional Data Distribution (Yellow & Blue Theme):



■ Fought (Yes) ■ No Fight (No)

Variable	Coef.	z-score	P> z
Intercept	-0.3339	-13.22	0.000
Helmet_Num	-0.6706	-16.15	0.000

3. SIGNIFICANCE & APPLICATION

MODEL STATISTICAL SIGNIFICANCE

OR = 0.5114

p-value = 2.086e-60 (<< 0.05)
95% CI: [0.471, 0.555]

CORE CONCEPT CONCLUSION

Through rigorous Logistic Regression Analysis on a nationwide youth sample, this study bridges physical regulatory law with behavioral psychology. The final results establish that institutional rule-following (helmet use) transcends simple safety; it mirrors an intrinsic risk-averse persona that strongly protects adolescents against societal aggression and physical confrontation.

FUTURE EXTENSIONS

- Behavioral Screening Radar: Schools can flag regions with low helmet compliance as early warning zones for potential youth aggression trends.
- Confounding Control: Introduce Multi-Variable Regression to adjust for family background variables to isolate pure personality clustering.

REJECT H0

With overwhelming statistical evidence ($p < 0.0001$, $Z = -16.15$), the null hypothesis is rejected. Adolescents who consistently wear helmets have 49% lower odds of participating in physical fights, strongly validating the "Safety-First Persona" behavioral clustering hypothesis.